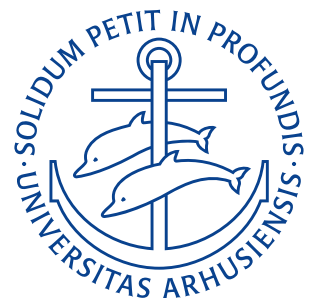


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Formula Sheet

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1 Constant Definitions

1.1 Coulomb Constant

$$k = \frac{1}{4\pi\epsilon_0} = 8,99 \cdot 10^9 \text{ Nm}^2/\text{C}^2.$$

Symbols

k	Coulomb constant	$8,99 \cdot 10^9 \text{ Nm}^2/\text{C}^2$
ϵ_0	Permittivity constant	$8,85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$

Lecture 1: Electrostatics

26. Januar 2026

2.1 Coulomb's Law

$$\mathbf{F} = k \frac{q_1 q_2}{r^2} \hat{r}.$$

Assumptions

- Point charges

Symbols

$\mathbf{F}$	Electrostatic force	N
k	Coulomb constant	$8,99 \cdot 10^9 \text{ Nm}^2/\text{C}^2$
q_1, q_2	Charges	C
r	Distance	m
$\hat{r}$	Unit vector	—

2.1.1 Principle of Superposition

$$\mathbf{F}_{1,\text{net}} = \sum_{i=1}^n \mathbf{F}_{1i}.$$

Symbols

$\mathbf{F}_{1,\text{net}}$	Net force	N
$\mathbf{F}_{1i}$	Force contribution on particle 1 from particle i	N

2.2 Quantization of Charge

$$q = \pm n \cdot e.$$

Symbols

q	Charge	C
n	Integer number ($n \in \mathbb{Z}$)	—
e	Elementary charge	$1,6 \cdot 10^{-19} \text{ C}$

2.3 Electric Field at a point

$$\mathbf{E}(x, y, z) = \frac{\mathbf{F}(x, y, z)}{q_0}.$$

Symbols

$\mathbf{E}(x, y, z)$	Electric field at point (x, y, z)	$\frac{\text{N}}{\text{C}}$
$\mathbf{F}(x, y, z)$	Force on test particle at point (x, y, z)	N
q_0	Test charge	C

2.3.1 Electric Field Produced by Point Charge

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{|q|}{r^2}.$$

Symbols

E	Electric field strength	$\frac{\text{N}}{\text{C}}$
ϵ_0	Permittivity constant	$8,85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$
q	Charge	C
r	Distance	m
$\hat{r}$	Unit vector	—

2.3.2 Electric Field due to a Dipole along dipole axis

$$E(z) = E_{(+)}(z) + E_{(-)}(z) = \frac{q}{4\pi\epsilon_0 z^2} \left(\frac{1}{1 - \frac{d^2}{2z}} - \frac{1}{1 + \frac{d^2}{2z}} \right) = \frac{q}{2\pi\epsilon_0 z^3} \left(\frac{d}{\left(1 - \left(\frac{d}{2z}\right)^2\right)^2} \right).$$

Symbols

E	Electric field strength	$\frac{\text{N}}{\text{C}}$
$E_{(+)}, E_{(-)}$	Contribution to E from the positive and negative part of the dipole	$\frac{\text{N}}{\text{C}}$
ϵ_0	Permittivity constant	$8,85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$
z	Distance along axis	m
d	Separation distance	m
q	Charge	C

2.3.3 Electric Field due to a Dipole along Dipole axis at Long Distance

$$E = \frac{1}{2\pi\epsilon_0} \frac{qd}{z^3} = \frac{1}{2\pi\epsilon_0} \frac{p}{z^3}.$$

Assumptions

- $z \gg d$, i.e. the distance between charges is much smaller than the distance from the charges you are measuring ("normal" case)

Symbols

E	Electric field strength	$\frac{\text{N}}{\text{C}}$
ϵ_0	Permittivity constant	$8,85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$
z	Distance along dipole axis	m
d	Separation distance	m
q	Charge	C
p	Electric dipole moment ($p = qd$)	Cm

2.3.4 Electric Field of a Charged Ring at Large Distance

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{z^2}.$$

Assumptions

- $z \gg R$, i.e. distance from the ring is much greater than the radius of the ring

Symbols

E	Electric field strength	$\frac{\text{N}}{\text{C}}$
ϵ_0	Permittivity constant	$8,85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$
q	Size of charge in the ring	C
z	Distance perpendicular to the radius of the ring	m

2.3.5 Electric Field of a Charged Disk

$$E = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{z}{\sqrt{z^2 + R^2}} \right).$$

Assumptions

- Infinitesimally thin disk

Symbols

E	Electric field strength	$\frac{\text{N}}{\text{C}}$
ϵ_0	Permittivity constant	$8,85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$
σ	Surface charge density	C/m^2
z	Distance perpendicular to the disk	m
R	radius of disk	m

2.3.6 Electric field from an infinite sheet

$$E = \frac{\sigma}{2\epsilon_0}.$$

Assumptions

- Infinitesimally thin infinite sheet

Symbols

E	Electric field strength	$\frac{\text{N}}{\text{C}}$
σ	Surface charge density	C/m^2
ϵ_0	Permittivity constant	$8,85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$

2.3.7 Electric field of a Conducting Sphere

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} = k \frac{q}{r^2}.$$

Assumptions

- Positive electric charge Q distributed uniformly throughout the volume of a conducting sphere.
- Spherical symmetry
- No charge enclosed in conductor

Symbols

E	Electric field strength	$\frac{\text{N}}{\text{C}}$
ϵ_0	Permittivity constant	$8,85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$
q	Charge	C
r	Distance	m
k	Coulomb constant	$8,99 \cdot 10^9 \text{ Nm}^2/\text{C}^2$

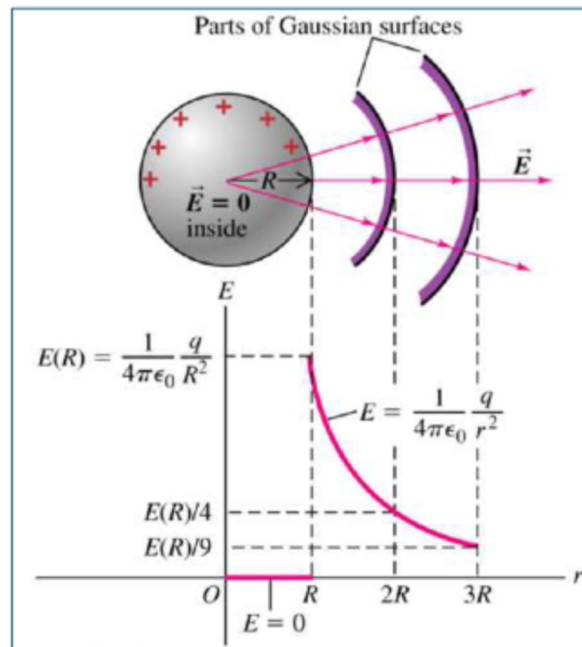


Figure 1: E-field propagation from a conducting sphere.

2.4 Electric Field of an Insulating Sphere

$$\begin{aligned} \text{Outside :} \quad E &= \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \\ \text{Inside :} \quad E &= \frac{q}{4\pi\epsilon_0 R^3} r. \end{aligned}$$

Assumptions

- Positive electric charge Q distributed uniformly throughout the volume of an insulating sphere.
- Spherical symmetry

Symbols

E	Electric field strength	$\frac{\text{N}}{\text{C}}$
r	Distance	m
R	Radius	m
ϵ_0	Permittivity constant	$8,85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$
q	Charge	C

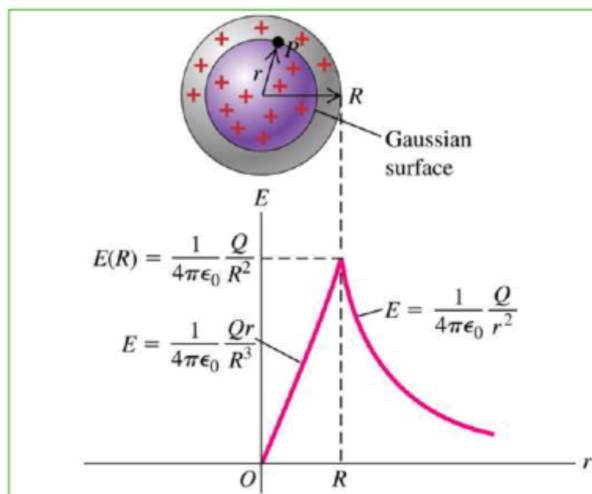


Figure 2: E-field propagation through an insulating sphere.

2.5 Gauss' Law

$$\Phi = \frac{q_{\text{enc}}}{\epsilon_0}.$$

Assumptions

- Vacuum (air)

Symbols

Φ	Electric flux	Vm
q_{enc}	Enclosed charge	C
ϵ_0	Permittivity constant	$8,85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$

2.5.1 Net electric flux through a surface caused by an electric field

$$\Phi = \oint E \cdot dA.$$

Symbols

Φ	Electric flux	Vm
E	Electric field vector	$\frac{\text{N}}{\text{C}}$
A	Area vector	m^2

2.6 Electric Potential Energy

$$U_E = q_0 E \Delta d.$$

Symbols

U_E	Electric potential energy	J
q_0	Test charge	C
E	Electric field strength	$\frac{\text{N}}{\text{C}}$
d	Separation distance	m

2.6.1 Electrostatic Potential

$$V_E = \frac{U_E}{q_0}.$$

Symbols

V_E	Electrostatic potential	V
U_E	Electric potential energy	J
q_0	Test charge	C

2.6.2 Electrostatic Potential of a Plate Capacitor

$$V_E = Ed.$$

Symbols

V_E	Electrostatic potential	V
E	Electric field strength	$\frac{\text{N}}{\text{C}}$
d	Separation distance	m

Lecture 2: Fundamentals of Electric Circuits

28. Januar 2026

3.1 Current, Voltage, and Resistance**3.1.1 Electric Current**

$$I = \frac{dq}{dt}.$$

Symbols

I	Current	A
q	Charge	C
t	Time	s

3.1.2 Electric Voltage

$$V = \frac{dW}{dQ}.$$

Symbols

V	Voltage	V
W	Work	J
Q	Total charge	C

3.1.3 Electric Resistance

$$R = \frac{V}{I}.$$

Symbols

R	Resistance	Ω
V	Voltage	V
I	Current	A

3.1.4 Current Density

$$J = \frac{I}{A}.$$

Symbols

J	Current density	A/m ²
I	Current	A
A	Cross-sectional area	m ²

3.1.5 Resistivity

$$\rho = \frac{E}{J}.$$

Symbols

ρ	Resistivity	$\Omega \text{ m}$
E	Electric field strength	$\frac{\text{N}}{\text{C}}$
J	Current density	A/m ²

3.1.6 Conductivity

$$\sigma = \frac{1}{\rho}.$$

Symbols

σ	Conductivity	$\Omega^{-1} \text{ m}^{-1}$
ρ	Resistivity	$\Omega \text{ m}$

3.1.7 Resistance From Resistivity

$$R = \frac{\rho L}{A}.$$

Symbols

R	Resistance	Ω
ρ	Resistivity	$\Omega \text{ m}$
L	Length	m
A	Cross-sectional area	m ²

3.1.8 Ohm's Law

$$I \propto V.$$

Symbols

I	Current	A
V	Voltage	V

3.1.9 Electric Power

$$P = VI = I^2 R = \frac{V^2}{R}.$$

Symbols

P	Electrical power	W
V	Voltage	V
I	Current	A
R	Resistance	Ω

3.2 Kirchoff's Circuit Laws**3.2.1 Kirchoff's Current Law (KCL)**

$$\sum_{n=1}^N I_n = 0.$$

Symbols

I_n	Current at node n	A
N	Amount of nodes	—

3.2.2 Kirchoff's Voltage Law (KVL)

$$\sum_{n=1}^N V_n = 0.$$

Symbols

V_n	Voltage difference between two points	V
N	Amount of points	N

3.2.3 Resistors in Series

$$R_{\text{EQ}} = \sum_{n=1}^N R_n.$$

Symbols

R_{eq}	Equivalent resistance	Ω
R_n	Resistance of resistor n	Ω
N	Amount of resistors in series	—

3.2.4 Voltage Divider Rule

$$V_n = \frac{R_n}{\sum_{n=1}^N R_n} V_s.$$

Symbols

V_n	Voltage difference over resistor n	V
R_n	Resistance over resistor n	Ω
V_s	Source voltage	V

3.2.5 Resistors in Parallel

$$R_{EQ} = \frac{1}{\sum_{n=1}^N \frac{1}{R_n}}.$$

Symbols

R_{eq}	Equivalent resistance	Ω
R_n	Resistance of resistor n	Ω
N	Amount of resistors in parallel	—

3.2.6 Current Divider Rule

$$I_n = \frac{\frac{1}{R_n}}{\sum_{n=1}^N \frac{1}{R_n}} I_s.$$

Symbols

I_n	Current difference over resistor n	A
R_n	Resistance over resistor n	Ω
I_s	Source current	A

Lecture 3: Principles of Circuit Analysis

2. Februar 2026

Lecture 4: Capacitors & Inductors

9. Februar 2026

5.1 Capacitors**5.1.1 Capacitance**

$$C = \frac{q}{V}.$$

Symbols

C	Capacitance	F
q	Charge	C
V	Voltage	V

5.1.2 Capacitance of a Parallel-Plate Capacitor

$$C = \epsilon_0 \frac{A}{d}.$$

Symbols

C	Capacitance	F
ϵ_0	Permittivity constant	$8,85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$
A	Cross-sectional area	m^2
d	Separation distance	m

5.1.3 I-V Relation for an Ideal Capacitor

$$I_C(t) = C \frac{dV_C(t)}{dt} \iff V_C(t) = \frac{1}{C} \int_{-\infty}^t I_C(t') dt'.$$

Symbols

I_C	Current through capacitor	A
V_C	Voltage over capacitor	V
C	Capacitance	F
t	Time	s

5.1.4 Capacitance of a Cylindrical Capacitor

$$C = 2\pi\epsilon_0 \frac{L}{\ln \frac{b}{a}}.$$

Symbols

C	Capacitance	F
ϵ_0	Permittivity constant	$8,85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$
L	Length	m
a	Radius of the inner post	m
b	Radius of the outer post	m

5.1.5 Capacitance of a Cylindrical Capacitor

$$C = 4\pi\epsilon_0 \frac{ab}{b-a}.$$

Symbols

C	Capacitance	F
ϵ_0	Permittivity constant	$8,85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$
a	Radius of the inner sphere	m
b	Radius of the outer sphere	m

5.1.6 Parallel Capacitance

$$C_{\text{eq}} = \sum_{i=1}^n C_i.$$

Symbols

n	Number of capacitors	—
C_{eq}	Equivalent capacitance	F
C_i	Capacitance of capacitor i	F

5.1.7 Capacitors in Series

$$\frac{1}{C_{\text{eq}}} = \sum_{i=1}^n \frac{1}{C_i}.$$

Symbols

n	Number of capacitors	—
C_{eq}	Equivalent capacitance	F
C_i	Capacitance of capacitor i	F

5.1.8 Potential Energy in a Capacitor

$$U_E = \frac{1}{2} CV^2.$$

Symbols

U_E	Electric potential energy	J
C	Capacitance	F
V	Voltage	V

5.2 Inductors**5.2.1 Inductance**

$$L = n \frac{\Phi_B}{I}.$$

Symbols

L	Inductance	H
n	Number of windings	—
Φ_B	Magnetic flux	T
I	Current	A

5.2.2 I-V Relation for an Ideal Inductor

$$V_L(t) = L \frac{dI_L(t)}{dt} \iff I_L(t) = \frac{1}{L} \int_{t_0}^t V_L(t') dt' + I_0.$$

Symbols

I_L	Current through inductor	A
V_L	Voltage over inductor	V
L	Inductance	H
t	Time	s

5.2.3 Magnetic Field of an Inductor

$$B = \mu_0 In.$$

Symbols

B	Magnetic field vector	T
-----	-----------------------	---

μ_0	Magnetic vacuum permeability	$1,26 \cdot 10^{-6} \text{ N/A}^2$
I	Current	A
n	Number of windings	—

5.2.4 Inductors in Series

$$L_{\text{eq}} = \sum_{i=1}^n L_i.$$

Symbols

n	Number of inductors	—
L_{eq}	Equivalent Inductance	H
L_i	Inductance of inductor i	H

5.2.5 Parallel Inductors

$$\frac{1}{L_{\text{eq}}} = \sum_{i=1}^n \frac{1}{L_i}.$$

Symbols

n	Number of inductors	—
L_{eq}	Equivalent Inductance	H
L_i	Inductance of inductor i	H

5.2.6 Energy Stored in an Inductor

$$U_L = \frac{1}{2} L I^2.$$

Symbols

U_L	Potential energy stored in inductor	J
L	Length	m
I	Current	A

5.2.7 Energy Density in an Inductor

$$u_L = \frac{B}{2\mu_0}.$$

Symbols

u_L	Energy density of inductor	J/m^3
B	Magnetic field vector	T
μ_0	Magnetic vacuum permeability	$1,26 \cdot 10^{-6} \text{ N/A}^2$

5.3 RC and RL Circuits in the DC Regime

5.3.1 Charge in a (Charging) Capacitor as a Function of Time

$$q(t) = C V_s \left(1 - e^{-\frac{t}{RC}} \right).$$

Symbols

q	Charge	C
C	Capacitance	F
V_s	Source voltage	V
t	Time	s
R	Resistance	Ω
C	Capacitance	F

5.3.2 Current in a (Charging) Capacitor as a Function of Time

$$I_C(t) = \frac{V_s}{R} e^{-\frac{t}{RC}}.$$

Symbols

I_C	Current through capacitor	A
C	Capacitance	F
V_s	Source voltage	V
t	Time	s
R	Resistance	Ω
C	Capacitance	F

5.3.3 Capacitive Time Constant

$$\tau_C = RC.$$

Symbols

τ_C	Capacitive Time Constant	s
R	Resistance	Ω
C	Capacitance	F

5.3.4 Inductive Time Constant

$$\tau_L = \frac{L}{R}.$$

Symbols

τ_L	Inductive Time Constant	s
L	Inductance	H
R	Resistance	Ω

5.3.5 Current in an Inductor

$$I = \frac{V_s}{R} \left(1 - e^{-\frac{t}{\tau_L}} \right).$$

Symbols

I	Current	A
V_s	Source voltage	V
R	Resistance	Ω

t	Time	s
τ_L	Inductive Time Constant	s